# University of Asia Pacific

**Department of Civil Engineering**

Course Code: **CE 202**

Course Name: **Engineering Materials Lab**


Course Teacher:

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## 1. Introduction:

The upcoming construction of a retail center in Rangpur demands exceptional quality in cement mortar and concrete. At UAP (University of Asia Pacific), extensive testing and analysis were carried out in order to satisfy this requirement. Evaluating various sand mixtures and combinations of fine and coarse particles was the main goal. To choose the best materials for the project, two crucial tests were used: sieve analysis and unit weight and void evaluation.

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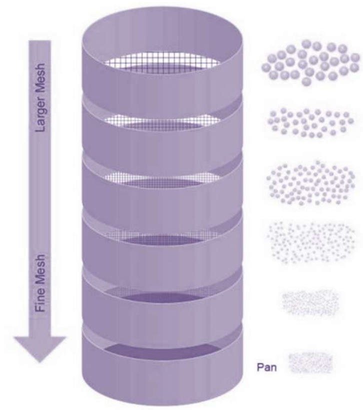



**Sieve Analysis:** Sieve Analysis: The term given to the sample operation of dividing a sample of aggregates into fraction each consisting of particles between specific limits. The analysis is conducted to determine the grading of material proposed for use as aggregates. The term fineness modulus (F.M) is a ready index of coarseness or fineness of material. It is an empirical factor obtained by adding the cumulative percentages of aggregates retained on each of the standard sieves and dividing this sum arbitrarily by 100.

No.100, No.50, No.30, No.16, No.8, No.4, 3/8 in.., ¾ in., 3 in.,1.5 in. are the ASTM standards sieves. This test method conforms to the ASTM standard requirements of specification C 136.

**Measurement of unit weight and voids:** This test method covers the determination of unit weight a compacted or loose condition of fine and coarse aggregates. Unit weight values of aggregates are necessary for use for many methods of selecting proportions for concrete mixtures. They may also be used for determining mass/volume relationships for conversions and calculating the percentages of voids in aggregates. Voids with him particles, either permeable or impermeable, are not included in voids as determined by this test method.

This test method conforms to the ASTM standard requirements of specification C29.

## 2. Project Background:

The decision to construct the retail center in Rangpur is a major development to meet the growing needs of people and the economy in the area. Success in such an endeavor does not depend on architectural design alone but also on the quality and durability of the construction material.

By conducting tests at UAP, we aimed to optimize the composition of cement mortar and concrete. This involved exploring various sand mixtures and combinations of fine and coarse aggregates through techniques such as Sieve Analysis and assessment of unit weight and voids. The goal is to identify the ideal material compositions that will ensure structural integrity, longevity, and aesthetic appeal for the retail center in Sylhet. This project exemplifies our commitment to delivering excellence in construction by leveraging scientific testing and analysis to inform our material selection and design decisions.

## 3. Methodology:

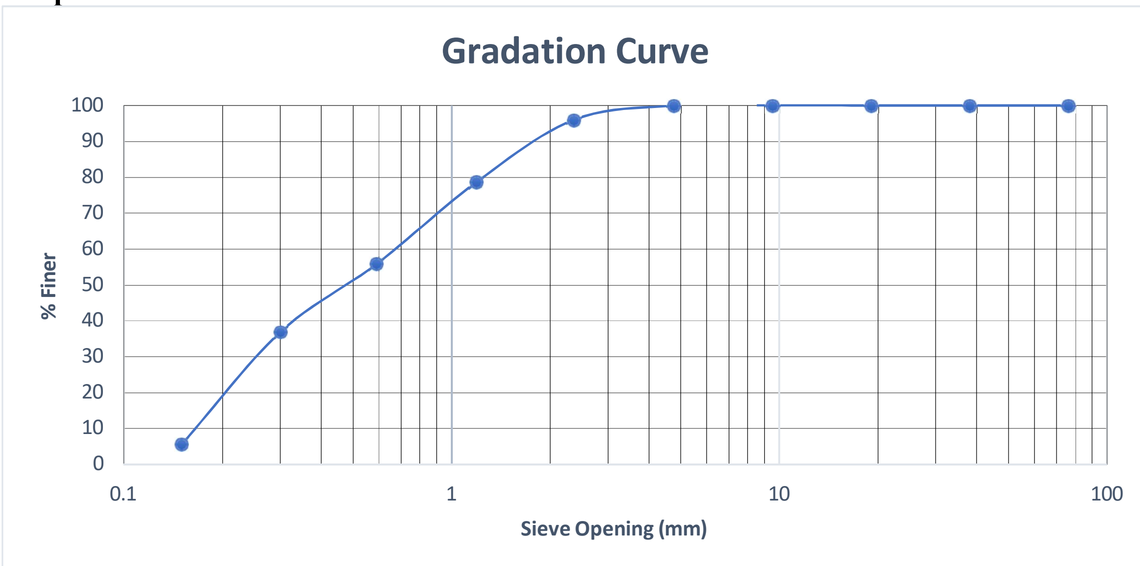
### i) Sieve Analysis and Fineness Modulus (FM) Calculation:

- Sand samples were collected based on the combinations specified in Table 1.
- Sieve analysis was performed on each sand sample to determine the particle size distribution.
- Fineness Modulus (FM) was calculated using the cumulative percentages retained on standard sieves.
- A graph was plotted using the sieve analysis data to determine the grading of each sand sample.

**Calculation:**

| Sieve Number | Sieve Opening (mm) | Materials Retained (gm) | % Materials Retained | Cumulative % Retained | % Finer |
|---|---|---|---|---|---|
| 3inch | 76.2 | 0 | 0 | 0 | 100 |
| 1.5 inch | 38.1 | 0 | 0 | 0 | 100 |
| 3/4 inch | 19.05 | 0 | 0 | 0 | 100 |
| 3/8 inch | 9.5 | 0 | 0 | 0 | 100 |
| #4 | 4.75 | 0.4 | 0.08 | 0.08 | 99.92 |
| #8 | 2.36 | 19.6 | 3.92 | 4 | 96 |
| #16 | 1.19 | 86.8 | 17.36 | 21.36 | 78.64 |
| #30 | 0.59 | 114 | 22.8 | 44.16 | 55.84 |
| #50 | 0.3 | 94.8 | 18.96 | 63.12 | 36.88 |
| #100 | 0.15 | 156.6 | 31.32 | 94.44 | 5.56 |
| Pan | - | 27.8 | 5.56 | 100 | 0 |
| Total | - | 500 | | | |
| FM | 2.2716 | | | | |

**Graph:**




The curve represents a well-graded distribution of fine aggregate particles. In a well-graded aggregate, there is a balanced mixture of various particle sizes within the material. This is evident in the graph where the curve appears smooth and relatively symmetrical, gradually increasing from left to right. The absence of steep slopes at either end of the curve indicates a well-balanced distribution without an excessive presence of very large or very small particles.

**ii) Unit Weight and Voids Test for Sand and Coarse Aggregate:**

- Unit weight and voids tests were conducted separately for the sand samples and coarse aggregates specified in Tables 1 and 2.
- The tests provided insights into the density and air voids present in the materials.

**For Sand Sample:**

| Test No | Test Method | T (kg) | G(kg) | V (m3) | M (kg/m3) | Avg. M (kg/m3) | Mssd (kg/m3) | % Void |
|---|---|---|---|---|---|---|---|---|
| 1 | Shoveling | 5.39 | 9.8 | 0.0028 | 1575.00 | 1577.38 | 1651.52 | 39.10 |
| 2 | | 5.39 | 9.82 | 0.0028 | 1582.14 | | | |
| 3 | | 5.39 | 9.8 | 0.0028 | 1575.00 | | | |
| 1 | Rodding | 5.39 | 10.11 | 0.0028 | 1685.71 | 1685.71 | 1764.94 | 34.91 |
| 2 | | 5.39 | 10.1 | 0.0028 | 1682.14 | | | |
| 3 | | 5.39 | 10.12 | 0.0028 | 1689.29 | | | |
| 1 | Jigging | 5.39 | 10.17 | 0.0028 | 1707.14 | 1708.33 | 1788.63 | 34.04 |
| 2 | | 5.39 | 10.19 | 0.0028 | 1714.29 | | | |
| 3 | | 5.39 | 10.16 | 0.0028 | 1703.57 | | | |

**For Coarse Aggregate:**

| Test No | Test Method | T (kg) | G(kg) | V (m3) | M (kg/m3) | Avg. M (kg/m3) | Mssd (kg/m3) | % Void |
|---|---|---|---|---|---|---|---|---|
| 1 | Shoveling | 10.42 | 23.94 | 0.00947 | 1427.67 | 1408.66 | 1474.87 | 40.31 |
| 2 | | 10.42 | 23.6 | 0.00947 | 1391.76 | | | |
| 3 | | 10.42 | 23.74 | 0.00947 | 1406.55 | | | |
| 1 | Rodding | 10.42 | 25.39 | 0.00947 | 1580.78 | 1584.65 | 1659.13 | 32.85 |
| 2 | | 10.42 | 25.41 | 0.00947 | 1582.89 | | | |
| 3 | | 10.42 | 25.48 | 0.00947 | 1590.29 | | | |
| 1 | Jigging | 10.42 | 26.2 | 0.00947 | 1666.31 | 1659.27 | 1737.26 | 29.69 |
| 2 | | 10.42 | 25.95 | 0.00947 | 1639.92 | | | |
| 3 | | 10.42 | 26.25 | 0.00947 | 1671.59 | | | |

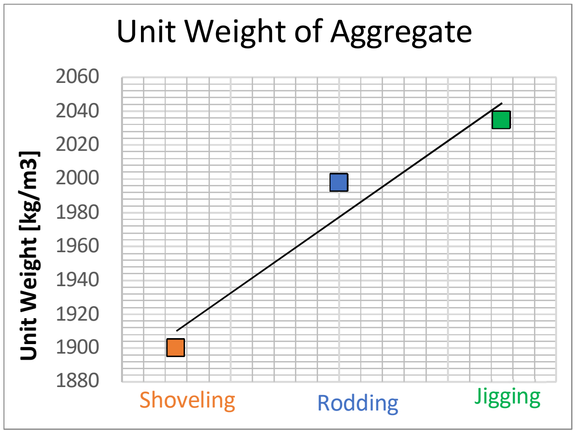
**iii) Combined Aggregate Test (Fine Aggregate and Coarse Aggregate):**

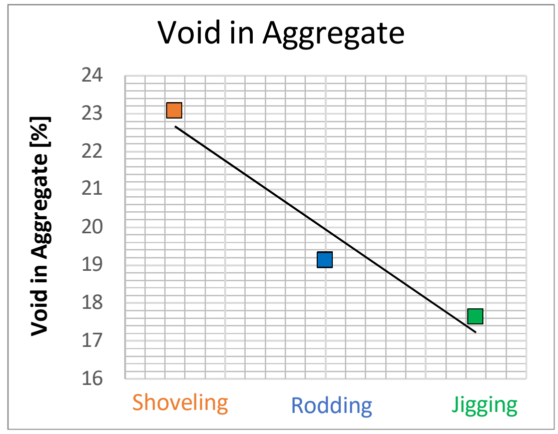
- Fine aggregate (FA) and coarse aggregate (CA) samples were combined in the specified ratios (FA:CA = 1.5) as per the group assignments.
- Graphs were plotted using the test data obtained for unit weights and voids in both scenarios.

**For Combined Aggregate:**

| Test No | Test Method | T (kg) | G(kg) | V (m3) | M (kg/m3) | Avg. M(kg/m3) | Mssd (kg/m3) | % Void |
|---|---|---|---|---|---|---|---|---|
| 1 | Shoveling | 5.39 | 10.66 | 0.0028 | 1882.14 | 1900.00 | 1989.30 | 23.08 |
| 2 | | 5.39 | 10.71 | 0.0028 | 1900.00 | | | |
| 3 | | 5.39 | 10.76 | 0.0028 | 1917.86 | | | |
| 1 | Rodding | 5.39 | 10.93 | 0.0028 | 1978.57 | 1997.62 | 2091.51 | 19.12 |
| 2 | | 5.39 | 11.04 | 0.0028 | 2017.86 | | | |
| 3 | | 5.39 | 10.98 | 0.0028 | 1996.43 | | | |
| 1 | Jigging | 5.39 | 11.08 | 0.0028 | 2032.14 | 2034.52 | 2130.15 | 17.63 |
| 2 | | 5.39 | 11.05 | 0.0028 | 2021.43 | | | |
| 3 | | 5.39 | 11.13 | 0.0028 | 2050.00 | | | |

**Graph for Unit Weights & Voids:**




- **% Void: Jigging < Rodding < Shoveling**
- **Less void → dense structure → lower porosity and permeability → higher strength → higher durability → economic**

## 4. Result Analysis and Discussion:

### 1. Fineness Modulus and Grading of Sand Combinations:

| Sand Combination, % | | Group | FM | Avarage FM | Grading |
|---|---|---|---|---|---|
| Local Sand | Sylhet Sand | | | | |
| 20 | 80 | 1 | 2.77 | 2.48 | Fine Sand |
| | | 3 | 1.97 | | |
| | | 5 | 2.7 | | |
| 40 | 60 | 2 | 2.43 | 2.14 | Fine Sand (possibly border line very Fine) |
| | | 4 | 1.71 | | |
| | | 6 | 2.27 | | |
| 60 | 40 | 7 | 1.92 | 2.17 | Fine Sand (possibly border line very Fine) |
| | | 9 | 2.39 | | |
| | | 11 | 2.19 | | |
| 80 | 20 | 8 | 2.42 | 1.94 | Very Fine Sand |
| | | 10 | 1.73 | | |
| | | 12 | 1.66 | | |

The FM values of the different sand mixes vary between 1.94 to 2.48, as can be seen from the table. This indicates that the sand mixes vary between fine and coarse. The highest FM value is for Sylhet sand. This is the coarsest sand in the table as per. With the lowest FM value, 1.94, the mix of 80% local and 20% Sylhet sand is the best sand in the table. The chart also shows the grading of the sand mixes. Grading is decided by the distribution of the particle sizes. The particle sizes in well-graded sand range from good large to microscopic. This is vital when making concrete as this affects the strength and durability of it. Sand with a graded gap contains no particles within a given size range. In the case of making strong concrete, this may be a problem. The grading of the sand combinations are not specified in the table, although some idea can be obtained from the FM values. For example, since Sylhet sand consists mainly of coarse particles, it is likely to be a poorly graded sand combination with the highest FM value. Since the FM value of the sand mixture (80% local sand, 20% Sylhet sand) is the lowest, it is probably well-graded because it contains mix particles.

**2. Unit weight and void for different sand sample:**

| Sand Combination, % | | Group | M (kg/m3) | Avg. M (kg/m3) | % Void | Avg. % Void |
|---|---|---|---|---|---|---|
| Local Sand | Sylhet Sand | | | | | |
| 20 | 80 | 1 | 1694 | 1665.67 | 35 | 35.67 |
| | | 3 | 1655 | | 36 | |
| | | 5 | 1648 | | 36 | |
| 40 | 60 | 2 | 1668 | 1685.67 | 36 | 35.00 |
| | | 4 | 1681 | | 35 | |
| | | 6 | 1708 | | 34 | |
| 60 | 40 | 7 | 1633 | 1649.00 | 37 | 36.33 |
| | | 9 | 1700 | | 34 | |
| | | 11 | 1614 | | 38 | |
| 80 | 20 | 8 | 1648 | 1616.33 | 36 | 37.33 |
| | | 10 | 1576 | | 39 | |
| | | 12 | 1625 | | 37 | |

The graph shows the variation in unit weight with the percentage of Sylhet and local sands. It can be observed from the graph that the unit weight increases proportionally with an increase in the fraction of Sylhet sand up to 60%. Beyond this 60%, further increases in the ratio of Sylhet sand lead to a decrease in unit weight. It also shows that the unit weight decreases with an increase in the fraction of local sand in the mix for a decrease in Sylhet sand.
In civil engineering, higher unit weights are always desired because higher unit weights increase the density of the sand, which is good for building.

According to the graph of voids percentage vs sand sample, the minimum percentage of voids belongs to 60% Sylhet sand and 40% local sand. For building, a small percentage of vacancy is good because then it requires less water in the mixture to mold, which enhances its strength.

Lower void content means that the aggregate particles are in closer contact with each other, and generally, the structure is stronger since the chances of flaws or cracks through the voids are reduced. In fact, with less space being occupied by air, less water is needed to fill up spaces between particles in the aggregate mix; thus, it provides a stronger bond for the aggregate and binding elements, such as cement in concrete.

**Unit weight and void for Coarse Aggregate:**

| Coarse Aggregate Size | Group | M (kg/m3) | Avg. M (kg/m3) | % Void | Avg. % Void |
|---|---|---|---|---|---|
| Greater than 3/4 inch | 1 | 1777 | 1684 | 25 | 28.67 |
| | 2 | 1734 | | 26 | |
| | 5 | 1669 | | 29 | |
| | 6 | 1659 | | 30 | |
| | 7 | 1628 | | 31 | |
| | 8 | 1637 | | 31 | |
| Less than 3/4 inch | 3 | 1769 | 1718.67 | 25 | 27 |
| | 4 | 1769 | | 25 | |
| | 9 | 1743 | | 26 | |
| | 10 | 1668 | | 29 | |
| | 11 | 1695 | | 28 | |
| | 12 | 1668 | | 29 | |

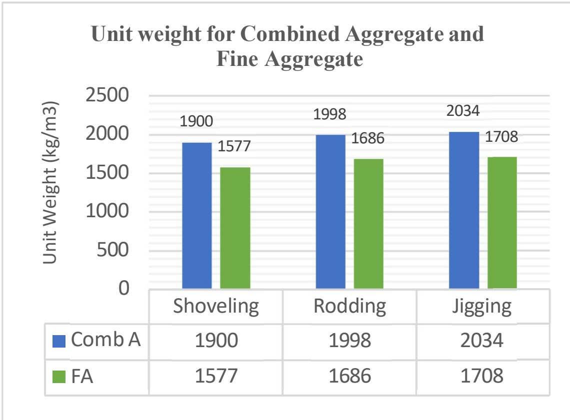
Unit weight is one of the important criteria in civil engineering with respect to material density and structural strength. Denser materials, which are represented by higher values of unit weight, are required to build strong, load-supporting structures that will withstand various forms of stresses and forces.

Voids percentage of a material are quite dependent upon the size of their coarse particles. Coarse aggregates used usually in concrete or asphalt mixtures typically comprise of pebbles and rock. Larger sizes of aggregates also have larger sized rocks within their mixture.

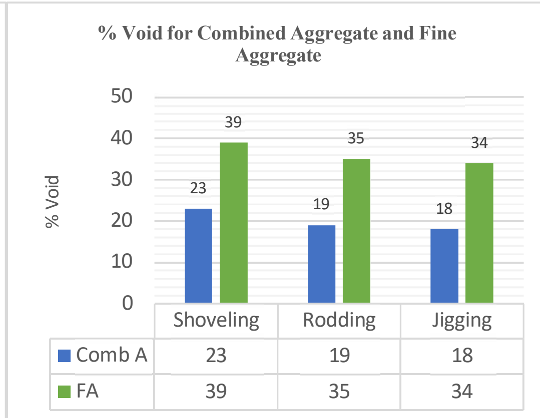
There are two different data series on this graph; these most likely can be referred to as aggregates blends or sources. Both series follow a trend, although without more context, it is hard to tell specifics about each series: as the size of the coarse aggregates increases, the unit weight generally decreases. This trend illustrates a fundamental concept in material science: since between the larger boulders, there are gaps or air spaces, they tend to have lower densities than the smaller ones.

**3. Unit weight and void for Combined Aggregate sample and Fine Aggregate sample:**

| Test Method | M (kg/m3) | | % Void | |
|---|---|---|---|---|
| | Combined Aggregate | Fine Aggregate | Combined Aggregate | Fine Aggregate |
| Shoveling | 1900 | 1577 | 23 | 39 |
| Rodding | 1998 | 1686 | 19 | 35 |
| Jigging | 2034 | 1708 | 18 | 34 |




| | Shoveling | Rodding | Jigging |
|---|---|---|---|
| Comb A | 1900 | 1998 | 2034 |
| FA | 1577 | 1686 | 1708 |




| | Shoveling | Rodding | Jigging |
|---|---|---|---|
| Comb A | 23 | 19 | 18 |
| FA | 39 | 35 | 34 |

***According to the Graph:***

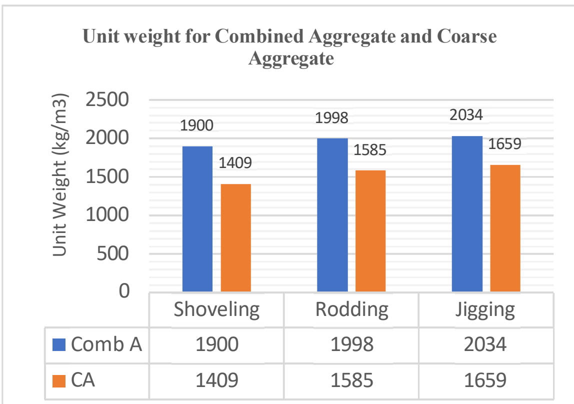
- **Shoveling:**
  - **Unit Weight:** Compared to the fine aggregate (1577 kg/m3), the blended aggregate has a larger unit weight (1990 kg/m3). This suggests that the fine aggregate alone is not as dense as the total aggregate.
  -
  - **Void Content:** The fine aggregate has a higher void percentage (39%), while the blended aggregate has a lower void content (23%). This supports the unit weight conclusion that the blended aggregate has less air spaces than the fine aggregate.

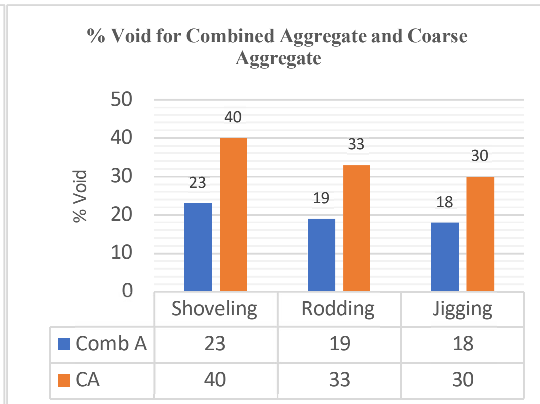
- **Rodding:**
  - **Unit Weight:**The unit weight of the mixed aggregate (1998 kg/m3) is greater than that of the fine aggregate (1686 kg/m3), much like shoveling.
  - **Void Content:** Once more, the fine aggregate (35%), in comparison to the mixed aggregate (19%), has a higher void content.
  -
- **Jigging:**
  - **Unit Weight:** According to the trend, the fine aggregate (1708 kg/m3) has a lower unit weight than the combined aggregate (2034 kg/m3)..
  - **Void Content:** Compared to the fine aggregate (34%), the mixed aggregate (18%) has less void content**.**

Overall, in all three methods, the data consistently show that the combined aggregate sample has lower void content and higher unit weight than the fine aggregate. That is because the combined aggregate fills up the spaces between the fine tiny aggregate particles. The combined aggregate mixes fine aggregate with coarse aggregate, normally made up of larger, denser particles. Consequently, the mixed aggregate has fewer air spaces between its particles and a heavier overall packing. It's interesting to note that optimal percentage of voids within aggregate may vary depending on their application in construction. However, lower void content is usually advantageous as it leads to a better, more durable, and impermeable end product.

**Unit weight and void for Combined Aggregate sample and Coarse Aggregate sample:**

| Test Method | M (kg/m3) | | % Void | |
|---|---|---|---|---|
| | Combined Aggregate | Coarse Aggregate | Combined Aggregate | Coarse Aggregate |
| Shoveling | 1900 | 1409 | 23 | 40 |
| Rodding | 1998 | 1585 | 19 | 33 |
| Jigging | 2034 | 1659 | 18 | 30 |




***According to the Graph:***

- **Shoveling:**
  - **Unit Weight:** At 1900 kg/m³, the amalgamated aggregate exhibits a significantly higher unit weight than the coarse aggregate, which weighs 1409 kg/m³. The aforementioned significant variation highlights the more compact arrangement attained via aggregation.
  - **Void Content:**Compared to the coarse aggregate, which has a 40% void content, the mixed aggregate has a significantly lower void content of 23%. This variation highlights the densification that was attained, suggesting a reduction in the number of air spaces in the combined material.
- **Rodding:**
  - **Unit Weight:** The combined aggregate has a higher unit weight of 1998 kg/m³ compared to the coarse aggregate's 1585 kg/m³, which is consistent with previous trends.
  - **Void Content:**Likewise, the mixed aggregate exhibits a lower void percentage (19%) as opposed to 33% for the coarse aggregate. The pattern of densification seen across various approaches is further supported by this uniformity.

- **Jigging:**
    - **Unit Weight:** This pattern continues in jigging, where the combined aggregate has a better unit weight (2034 kg/m³) than the coarse aggregate (1659 kg/m³).
    - **Void Content:** In accordance with this, the combined aggregate exhibits a lower void content—18% as opposed to 30% for the coarse aggregate. This consistency supports the general densification tendency that has been seen in all approaches.

Reductions in void content are typically associated with improved strength, durability, and impermeability in the finished product.

- **Conclusion:**

According to the test results, the cement mortar and concrete that will provide balanced particle sizes, proper density, and low void content in the Sylhet retail center project are determined. Such elements will ensure the building is strong, stable, and durable as required. These findings could considerably help in the selection and mixing of concrete mixtures to project requirements. The above explanation elaborates on the aggregate with combined unit weight and less permeable voids. In order for the concrete to have the best strength and durability, the combined aggregate must be prepared using the jigging method. Overall, group 2's aggregated aggregate is the best for the project based on its highest unit weight and least permeable void.